Gradient Activation of Syntactic Structures Facilitates Recovery from Garden Paths

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Garden-path sentences such as "The brave soldier captured // by the enemy had memorized the route" temporarily allow two analyses: the "soldier" can be interpreted as either the agent or patient of "captured". Previous work primarily focused on the reanalysis behavior at the disambiguating region [cf. 1], but less is known about how readers' early commitments to potential readings vary among individuals and how that impacts their revision performance. One possibility is that early graded commitments may permit more flexible updating when disambiguating input arrives [2-3]: whereas "categorical" readers strongly commit to a single parse, "gradient" readers maintain partial commitments to several alternatives and may therefore reconstruct the correct parse more efficiently. This study asks (1) whether gradient structural activation facilitates recovery from garden paths, and (2) whether this facilitation, if any, is modulated by the actual parse that readers initially prefer and their sensitivity to structural probabilities that guide commitment distribution.

[Methods] Exp1 (Slider Task). Participants (N=31) read garden-path sentence fragments truncated before the disambiguation words. We included 40 items from two ambiguity types: Matrix Verb/Reduced Relative Clause (MV/RR) & Noun Phrase/Sentential Complement (NP/S). We manipulated the structural probability of these fragments to form a continuum ranging from 100% likely to receive MV (/NP) readings to 100% likely to get RR (/S) readings. Structural probabilities were estimated from GPT-2 sentence completions. Each fragment appeared with a slider and two candidate next words. Participants indicated their preferred continuation by positioning the slider, reflecting their structural interpretation. For each participant, we derived: (1) Structural Parsing Gradiency, quantified as the quadratic coefficient from a regression model fit to the distribution of slider click locations, where smaller coefficients indicate more gradient commitments to potential parses; (2) Initial Bias, computed as the mean slider value for each item across three repetitions; and (3) Structural Sensitivity, estimated as the by-participant slope from a linear mixed-effects model predicting slider responses from structural probability. **Exp2** One week later, the same participants read full versions of the sentence fragments with RR and S structures in a **bidirectional self-paced reading task [4].** Ambiguity was manipulated (garden paths vs. control). The disambiguating word served as the critical region (CR). The first-pass reading times and regressive rereading times (FPRT and RegRRT) were analyzed at the subject NP, pre-critical (CR-1), CR, and first spillover (CR+1) regions.

[Results] ***Ambiguity Effects*** [MVRR] FPRT showed Ambiguity effects at CR-1. [NP/S] FPRT revealed Ambiguity effects at CR-1, CR and CR+1. RegRRT showed effects at CR-1 and CR. ***Individual Trait Predictors*** [MV/RR] FPRT (1) CR-1: Linear mixed-effects models revealed a main effect of Gradiency from the slider tasks on the ambiguity effects ($p<.05$). Gradient readers showed larger ambiguity effects than categorical readers. (2) CR: Gradiency ($p<.01$) and Initial Bias ($p<.05$) predicted reading times in the garden-path condition: gradient and RR-biased readers showed shorter reading times than categorical and MV-biased readers, respectively. (3) CR+1: There were main effects of Gradiency ($p<.05$) and Structural Sensitivity ($p<.01$). Gradient and sensitive readers showed smaller ambiguity effects than categorical and insensitive readers. [NP/S] RegRRT at the subject NP region Gradiency ($p<.05$) and its interaction with Initial Bias ($p<.05$). Gradient readers showed larger ambiguity effects than categorical ones – the difference was reduced when they were initially NP-biased.

[Conclusions] When ambiguities first arise, maintaining multiple readings requires extra effort. Crucially, this investment facilitates later recovery from the garden paths once the actual parse is revealed. Recovery is further aided when a reader's initial bias aligns with the target parse and when they are sensitive to probabilistic contextual cues that guide structural analysis. This pattern robustly characterizes MV/RR items, but not NP/S items, where gradient readers instead show active, confirmatory rereading rather than clear evidence of facilitated recovery [5]. Ongoing work tests cross-structural differences and examines whether structural parsing gradiency correlates with speech perception gradiency to explore potential domain-generality [6].

Table 1. Stimuli. CR = critical region, GP = garden paths.

MV/RR (Matrix Verb/Reduced Relative Clause)

EXP1 (sentence fragments)		The brave soldier captured...				MV reading: e.g., ...the terrorist. RR reading: e.g., ... by the enemy had..			
		Subject		CR-1	CR	CR+1			
EXP2	GP (target parse: RR)	The brave soldier		captured	**by**	the enemy	had	memorized	the route.
	control	The brave soldier	*who was*	captured	**by**	the enemy	had	memorized	the route.

NP/S (Noun Phrase/Sentential Complement)

EXP1 (sentence fragments)		The beautiful guitarist knew the song...			NP reading: e.g., ...roughly. S reading: e.g., ...would move...			
		Subject		CR-1	CR	CR+1		
EXP2	GP (target parse: S)	The beautiful guitarist	knew	the song	**would**	move	the audience.	
	control	The beautiful guitarist	knew *that*	the song	**would**	move	the audience.	

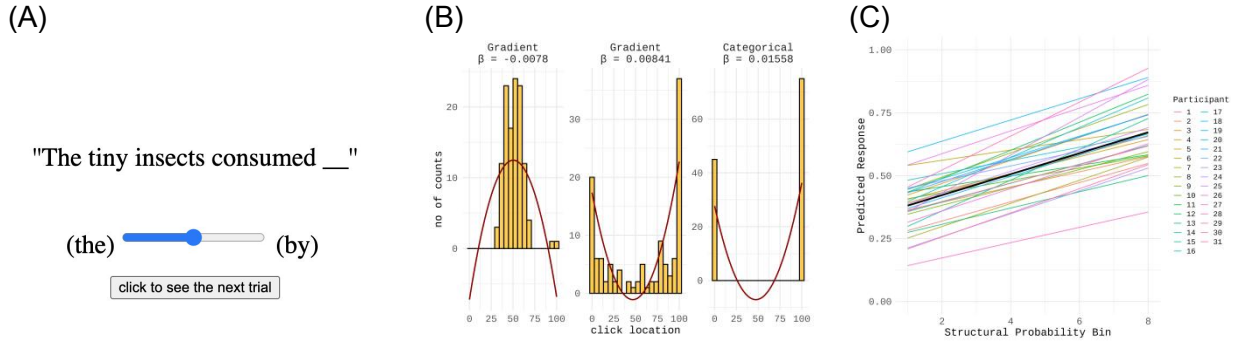

Fig. 1 **Visual Display and Results of EXP1.** (A) Example trial from the slider task. (B) Distributions of slider click locations for three representative readers. Red curves show the quadratic fit. Steeper curves (higher β) indicate more categorical parsing. (C) Predicted slider responses across structural probability bins. Steeper lines (larger slopes) reflect greater sensitivity to variations in structural probability.

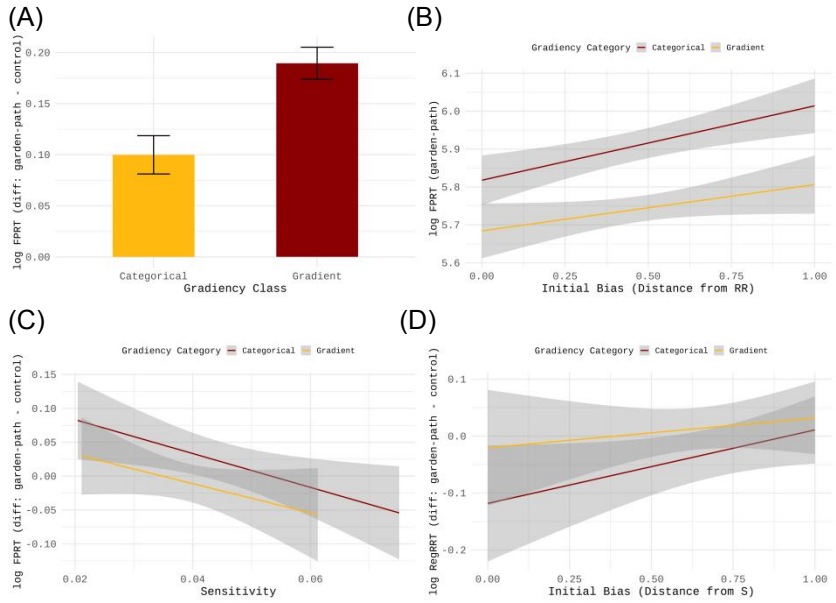

Fig. 2 **Combined Results of EXP1 & 2.** For visualization, participants were grouped into gradient (red) and categorical (yellow) readers using the median Gradiency score as the cutoff. Panels A-C show MV/RR results; Panel D shows NP/S. (A) Ambiguity effects by Gradiency at CR-1 ("captured"). (B) Reading times at CR ("by") in the garden-path condition predicted by Gradiency and Initial Bias. (C) Ambiguity effects by Gradiency and Sensitivity at CR+1 ("the enemy"). (D) Ambiguity effects by Gradiency and Initial Bias at the subject NP ("The beautiful guitarist").

References: [1] MacDonald et al. (1994). *Psychological Review*. [2] Kapnoula et al. (2021). *Journal of Experimental Psychology. Human Perception and Performance*. [3] Gwilliams et al. (2018). *The Journal of Neuroscience*. [4] Paape & Vasishth. (2022a). *Cognitive Science*. [5] Paape & Vasishth. (2022b). *Glossa Psycholinguistics*. [6] Kong & Edwards. (2016). *Journal of Phonetics*.